QC Inspection Station — Ladder Log

Rung	Section	Section Name	Input Contacts
1	A	Pen fill → cycle start	NO: DI_02 (pen full — 5-6 pcs) NO: M0 (seal-in) NC: CTU_1.Q (batch done)
2	A	Pen fill — mid alert	NC: DI_01 (mid sensor LOW → fewer than 3 pcs) NO: M0 (cycle running)
3	B	Gate open — one-shot	NO: M0 (cycle run) NC: M1 (reject in progress) NC: TON_1.Q (gate-dwell done)
4	B	Gate dwell timer	NO: DO_00 (gate open)
5	C	Inspection trigger	NO: DO_00 (gate just opened → rising edge) NC: TON_2.Q (inspect done)
6	C	Good part count	NO: TON_2.Q (within inspect window) NO: DI_03 (sensor ON = good)
7	C	Bad part detection — set reject flag	NO: TON_2.Q NC: DI_03 (sensor OFF = bad)
8	C	Total counter (all parts)	NO: TON_2.Q (inspect window)
9	D	Reject cylinder — extend	NO: M1 (reject flag set)
10	D	Reject dwell / reset flag	NO: M1 NO: TON_3.Q (dwell complete)
11	E	Green RUN light	NO: M0 (cycle running)

12	E	Red DONE / BAD light	NO: CTU_1.Q (batch done) NO: M1 (bad part) NO: TOF_1.Q (5 s hold)
13	E	Pen refill flash indicator	NC: DI_00 (low-level sensor — pen empty/low) NO: TOF_2.Q (500 ms blink clock)
14	F	HMI totals display	NO: M0 (always live while running)
15	F	Counter reset (HMI button)	NO: HMI_RST (HMI reset coil bit)

	S
A	Pen Fill / Cycle Start
B	Gate Control
C	Inspection & Count
D	Sort / Reject
E	Status Lights
F	HMI & Motor

ic Rung Summary (Rev B — 3 Feed

Coil / Output	Timer / Counter	Preset
M0 (Cycle Run flag)		
DO_06 (Amber 'Low-mid' indicator)		
DO_00 (Pneumatic gate solenoid)	TON_1 — gate dwell	0.6 s
TON_1 (Gate-dwell timer)	TON_1	0.6 s
TON_2 (Inspection delay timer)	TON_2	0.4 s
CTU_1 (Good-parts counter) + CTU_2 (Total counter)	CTU_1 / CTU_2	CTU_1 PV = batch size
SET M1 (Reject flag)		
CTU_2 (Total parts counter)	CTU_2	—
DO_01 (Reject cylinder solenoid)	TON_3 — reject dwell	1.0 s
RST M1 (clear reject flag)	TON_3	1.0 s
DO_07 (Green light)		

DO_08 (Red light) + TOF_1 (hold timer)	TOF_1	5.0 s
DO_09 (Amber flash lamp)	TOF_2 (blink)	0.5 s
AO_00 (Motor speed) via HMI register HMI: Total, Good, Bad, FPY, Reset btn		
RST CTU_1 / RST CTU_2		

ection Color Legend

l-Pen Sensors)

Logic & Notes
DI_02 (full-level sensor) or M0 seal latches cycle ON. CTU_1.Q (count reached) breaks the latch and stops the cycle.
If mid sensor is not seeing product while cycle is running, amber light warns operator to prep a refill soon. Non-stopping alert.
Gate retracts to release one piece. NC M1 interlocks gate closed during active rejection. Gate stays open 0.6 s then TON_1.Q NC breaks it closed.
Starts when gate opens. Q-bit feeds NC contact in rung 3 to close gate after dwell.
Fires inspection window 0.4 s after gate opens (transit time). Q ends the window.
Both counters increment on each good part. CTU_1.Q stops cycle (rung 1 NC).
If part is bad during inspect window, M1 is latched ON to activate reject cylinder.
Increments for every part (good and bad) — used in FPY = CTU_1 / CTU_2 on HMI.
Cylinder extends to push bad part off conveyor. Holds for 1 s.
After 1 s dwell TON_3.Q rises, RST M1 clears the reject flag and retracts cylinder.
Green on while cycle is active.

Red light: on when batch complete OR bad part. TOF_1 holds it for 5 s after bad-part flag clears.
When pen is empty (DI_00 OFF) the blink timer gates the lamp for a visible flash. TOF_2 re-triggers on its own Q falling edge.
HMI reads CTU_1.CV (good), CTU_2.CV (total). FPY = CTU_1.CV / CTU_2.CV calculated in HMI. Reset button clears both counters.
HMI 'Reset' button triggers HMI_RST momentary bit, clearing both counters and FPY.

QC Stat		
Tag	**Type**	**I/O Class**
DI_00	BOOL	Digital IN
DI_01	BOOL	Digital IN
DI_02	BOOL	Digital IN
DI_03	BOOL	Digital IN
DI_04	BOOL	Digital IN
HMI_RST	BOOL	HMI Bit
DO_00	BOOL	Digital OUT
DO_01	BOOL	Digital OUT
DO_06	BOOL	Digital OUT
DO_07	BOOL	Digital OUT
DO_08	BOOL	Digital OUT
DO_09	BOOL	Digital OUT
AO_00	INT	Analog OUT
M0	BOOL	Internal
M1	BOOL	Internal
TON_1	TON	Timer
TON_2	TON	Timer
TON_3	TON	Timer
TOF_1	TOF	Timer
TOF_2	TOF	Timer
CTU_1	CTU	Counter
CTU_2	CTU	Counter

ion — Tag Register

Description
Feed Pen Proximity Sensor 1 — LOW LEVEL (OFF = pen empty → flash alert)
Feed Pen Proximity Sensor 2 — MID LEVEL (ON = ≥3 pcs present; OFF = low-mid warning)
Feed Pen Proximity Sensor 3 — FULL LEVEL (ON = 5-6 pcs; enables cycle start)
QC Inspection Sensor — ON = part is GOOD, OFF = part is BAD
Light Curtain / Gate Confirmation — part passed through sort area
HMI Reset button — momentary coil bit; resets CTU_1 and CTU_2
Pneumatic Gate Solenoid — RETRACT = release one piece to inspection
Reject Cylinder Solenoid — EXTEND = push bad part off conveyor
Amber 'Low-Mid' Indicator — lit when mid sensor is not seeing product during run
Green RUN light — ON while cycle active (M0)
Red DONE/BAD light — ON when batch complete or bad part; 5 s hold via TOF_1
Amber FLASH lamp — blinks when pen is empty (DI_00 OFF); 500 ms blink via TOF_2
Motor Speed Reference — set from HMI register (0–100 %)
Cycle Run flag — sealed by rung 1; cleared by CTU_1.Q when batch count reached
Reject flag — SET by rung 7 (bad part); RST by rung 10 after 1 s dwell
Gate dwell timer — PT = 0.6 s; Q closes gate after part is released
Inspection delay timer — PT = 0.4 s; defines inspection window after gate opens
Reject dwell timer — PT = 1.0 s; holds cylinder extended; Q resets M1
Red-light hold timer — PT = 5.0 s off-delay; keeps red light on after bad part clears
Blink clock — PT = 0.5 s; creates flash pattern for pen-refill lamp when DI_00 is OFF
Good-parts counter — CV = good count; Q = batch done (CV ≥ PV); PV set per batch size
Total-parts counter — CV = all parts; used in FPY = CTU_1.CV / CTU_2.CV on HMI

QC Station — Operational Sequence & Timer	
Step	**Description**
1	Operator fills feed pen to at least 5–6 pieces. DI_02 (full sensor) turns ON.
2	DI_02 energizes rung 1, M0 (cycle run flag) seals in. Green light (DO_07) turns ON. If mid sensor (DI_01) is also ON — no amber light.
2a	If pen drops below mid level while running: DI_01 goes OFF → DO_06 amber 'Low-Mid' lamp lights. Cycle continues; operator preps refill.
3	Gate (DO_00) retracts — releases one piece to inspection area. TON_1 gate-dwell timer starts (0.6 s).
4	After 0.6 s TON_1.Q NC breaks rung 3 → gate closes.
5	TON_2 inspection delay fires (0.4 s) to allow part transit to sensor.
6a	TON_2.Q + DI_03 ON → GOOD part: CTU_1 (good) and CTU_2 (total) both increment.
6b	TON_2.Q + DI_03 OFF → BAD part: SET M1 (reject flag). CTU_2 (total) increments.
7	M1 energizes DO_01 (reject cylinder) — extends to push bad part off. Red light (DO_08) turns ON. TOF_1 5 s hold starts.
8	After 1.0 s TON_3.Q → RST M1, cylinder retracts. Red light stays on for 5 s total (TOF_1 hold).
9	Cycle repeats from step 3 for each piece until CTU_1.CV ≥ PV (batch done).
10	CTU_1.Q NC breaks rung 1 → M0 drops → cycle stops. Green light OFF. Red light ON (batch done).
11	If pen is empty (DI_00 OFF): DO_09 amber lamp flashes at 0.5 s rate via TOF_2 blink clock.
12	HMI displays: Total (CTU_2.CV), Good (CTU_1.CV), Bad (CTU_2−CTU_1), FPY = Good/Total.
13	Operator presses HMI Reset → HMI_RST bit → RST CTU_1 and CTU_2 → FPY clears.

Timer & Counter Preset Reference	
Tag	**Preset (PT / PV)**
TON_1	PT = 0.6 s
TON_2	PT = 0.4 s
TON_3	PT = 1.0 s
TOF_1	PT = 5.0 s
TOF_2	PT = 0.5 s

CTU_1	PV = batch size
CTU_2	PV = —

Presets (Rev B)

Controlled By
DI_02 (physical sensor)
Rung 1, 2, 11
Rung 2, DO_06
Rung 3, TON_1
TON_1.Q → Rung 3
Rung 5, TON_2
Rung 6, CTU_1, CTU_2
Rung 7, 8, M1
Rungs 9, 12
Rung 10, TON_3, TOF_1
Rung 1 NC CTU_1.Q
Rung 1, 11, 12
Rung 13, TOF_2
HMI register / Rung 14
Rung 15

Purpose
Gate dwell — holds gate open long enough for one piece to clear
Inspection window delay — transit time from gate to sensor
Reject dwell — cylinder extension time to clear bad part
Red-light hold after bad part / batch complete
Pen-refill blink clock — 0.5 s period flash

Good-parts count; Q stops cycle when CV ≥ PV
Total-parts count; used for FPY calculation on HMI